

The Instruments of Trade Policy

ECON 308 – International Economic Relations Chapter 9

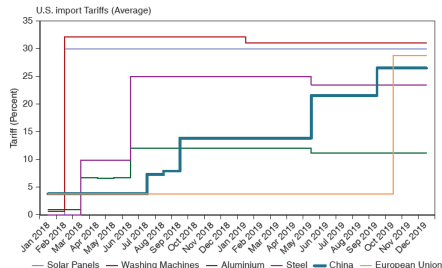
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Opening Case: The Canada-US Steel Tariff Dispute (2018)

- June 2018: US imposed 25% tariff on Canadian steel and 10% on aluminium under “national security” (Section 232).
- Canada retaliated with C\$12.6 billion in counter-tariffs on US goods (steel, aluminium, orange juice, ketchup, bourbon).
- US steel prices rose $\sim 25\%$; US car and appliance manufacturers protested higher costs.
- The tariffs were removed in May 2019, but cost both economies real resources during the dispute.
- **This chapter provides the tools to analyse who paid, who gained, and how much was lost.**



Learning Objectives

1. Distinguish specific from ad valorem tariffs and calculate their effects.
2. Use consumer and producer surplus analysis to decompose tariff welfare effects.
3. Derive the optimal tariff for a large country.
4. Compare tariffs and import quotas as policy instruments.
5. Evaluate the welfare effects of export subsidies.
6. Apply these tools to real US/Canadian trade policy episodes.

Types of Trade Policy Instruments

Tariffs:

- **Specific tariff:** fixed dollar amount per unit (e.g., \$0.25/kg of steel).

$$P_d = P_w + t$$

- **Ad valorem tariff:** percentage of import value (e.g., 25% on steel).

$$P_d = P_w(1 + \tau)$$

- **Tariff-rate quota (TRQ):** low tariff up to a quota; high tariff above.

Non-tariff barriers:

- **Import quota:** physical limit on imports.
- **Voluntary export restraint (VER):** foreign

All variables:

P_w = world price (exogenous for small country)

$P_d = P_w + t$ = domestic price after tariff

t = specific tariff rate

τ = ad valorem tariff rate

Q_S = domestic supply

Q_D = domestic demand

$M = Q_D - Q_S$ = imports

CS = consumer surplus

PS = producer surplus

GP = government revenue from tariff

Home Import Demand and Foreign Export Supply

Deriving import demand (MD):

- At low prices: domestic demand $\gg$ supply $\Rightarrow$ large imports.
- As price rises: domestic supply rises and demand falls $\Rightarrow$ imports fall.
- $MD = Q_D(P) - Q_S(P)$: downward-sloping in price.

Foreign export supply (XS):

- At low prices: foreign industry produces little, so few exports.
- As price rises: foreign supply rises $\Rightarrow$ more exports.
- XS: upward-sloping in price.

World equilibrium: $MD = XS$ at P_w .

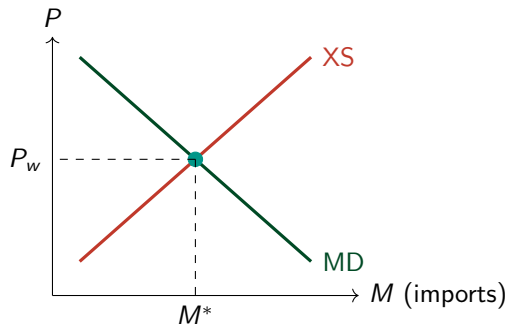
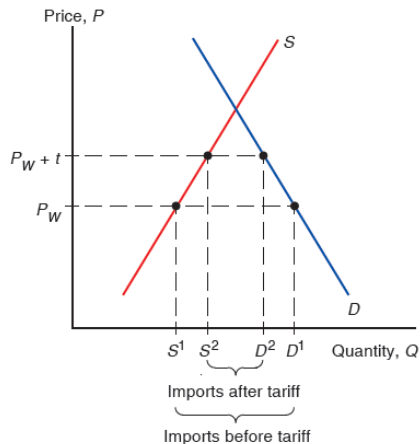


Figure 9-5: A Tariff in a Small Country



What this figure shows:

- **Small country:** cannot affect world price P_W .
- A specific tariff t raises the domestic price to $P_W + t$.
- Domestic production rises from Q_1 to Q_2 (higher price incentivises producers).
- Domestic consumption falls from Q_4 to Q_3 (higher price discourages consumers).
- Imports fall from $(Q_4 - Q_1)$ to $(Q_3 - Q_2)$.
- **Four areas:** a (PS gain), b (production DWL), c (tariff revenue), d (consumption DWL).

Consumer Surplus: The Foundation of Welfare Analysis

Consumer surplus (CS): the difference between what consumers are *willing to pay* and what they *actually pay*.

$$CS = \int_0^{Q_D} D^{-1}(Q) dQ - P \cdot Q_D$$

Graphically: area below the demand curve and above the price.

When price rises by ΔP :

$$\Delta CS \approx -Q_D \cdot \Delta P - \frac{1}{2} \Delta Q_D \cdot \Delta P$$

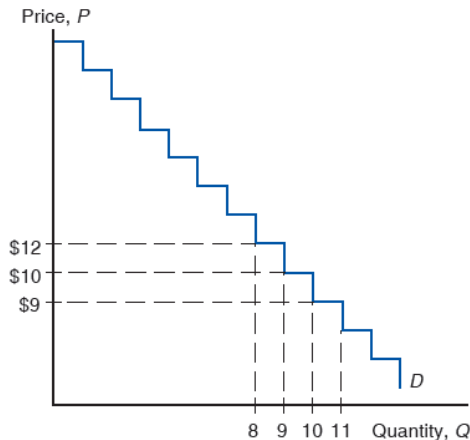
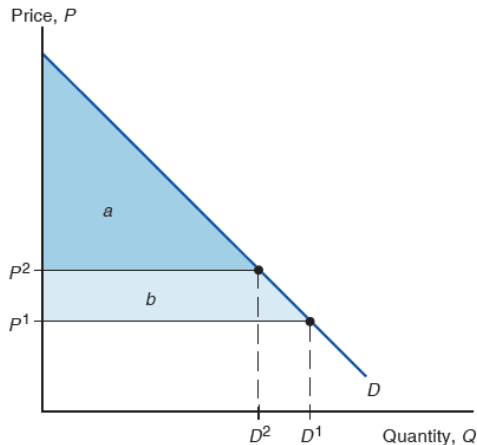


Figure 9-6: Geometry of Consumer Surplus



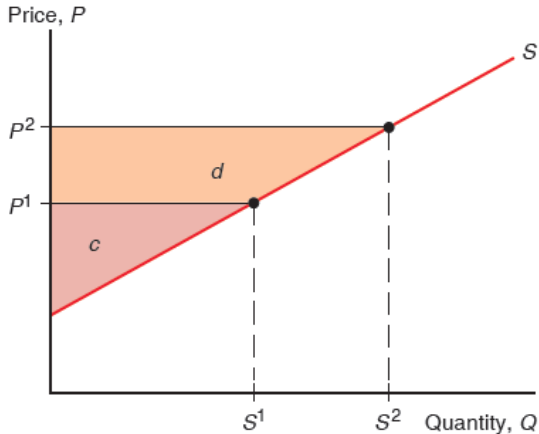
Reading Figure 9-6:

- CS is the shaded area between the demand curve and the price line.
- At a **lower price**: CS is the large triangle above the low price and below the demand curve.
- At a **higher price**: CS shrinks to the small triangle above the high price.
- The **loss in CS** from a price rise = the trapezoid between the two price levels.
- This is exactly the area $(a + b + c + d)$ in the tariff welfare diagram.

Key message

Consumers always lose when prices rise from a tariff, and the loss is proportional to the price increase and the

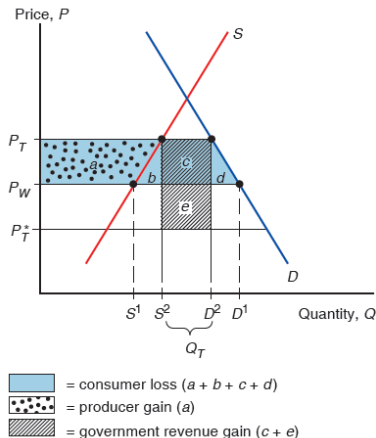
Figure 9-8: Geometry of Producer Surplus



Reading Figure 9-8:

- **Producer surplus (PS):** area above the supply curve and below the price.
- PS represents *profit* and rents to factors in the protected industry.
- When price rises from P_w to $P_w + t$: PS increases by the area between the two prices and above the supply curve (area *a*).
- This is a *transfer* from consumers to producers, not a net gain for society.
- But: concentrated PS gain motivates industry lobbying for protection.

Figure 9-9: Costs and Benefits of a Tariff for the Importing Country



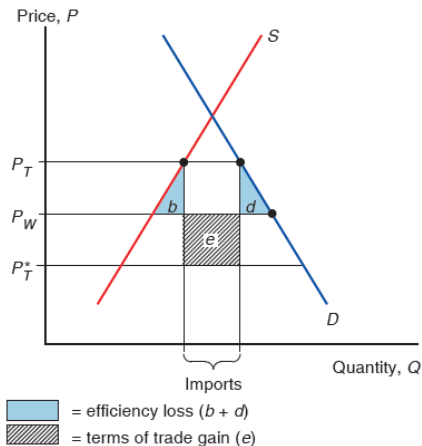
Decomposing tariff welfare effects:

Group	Area	Effect
Consumers	$-(a + b + c + d)$	Lose
Producers	$+a$	Gain
Government	$+c$	Revenue
Net (small)	$-(b + d)$	Lose

Area b : production distortion — resources wasted in inefficient domestic production that could have been imported more cheaply.

Area d : consumption distortion — consumers priced out of goods they would have bought at world prices.

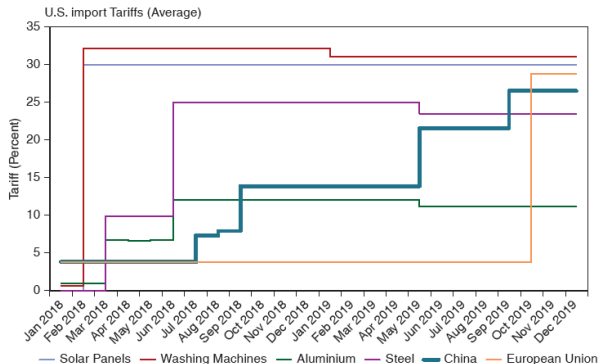
Figure 9-10: Net Welfare Effects of a Tariff



What Figure 9-10 shows:

- Consolidates the welfare analysis: net welfare change = $-(b + d)$ for a small country.
- **Both triangles are losses:** they represent resources misallocated due to the price distortion.
- The size of the deadweight loss (DWL) depends on:
 - The tariff rate (higher tariff $\Rightarrow$ bigger DWL).
 - The price elasticities of supply and demand (more elastic $\Rightarrow$ bigger $b + d$).
- For a *large country*: there is also a ToT gain (area e) if world price falls. Net = $e - (b + d)$.

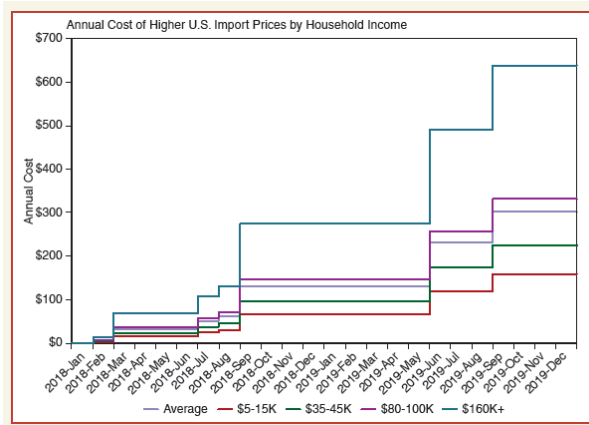
Figure 9-11: Average Tariffs on US Imports, 1900–2023



Historical perspective:

- US tariffs averaged $>40\%$ in the 1930s (Smoot-Hawley, 1930).
- Smoot-Hawley triggered global retaliation: world trade collapsed 66% (1929–1933).
- GATT rounds progressively reduced tariffs:
 - 1947–1994: average fell from $\sim 40\%$ to $<5\%$.
 - Uruguay Round (1995): WTO, $<4\%$ average.
- **2018 Trump tariffs:** temporary reversal — Section 232 (steel/aluminium), Section 301 (China).
- Average US tariff remains $<5\%$ but

Figure 9-12: Annual Cost of Higher Prices by Household Income



Who bears the cost of tariffs?

- This figure shows the annual cost of US tariff-induced price increases by household income quintile.
- **Key finding:** tariffs are *regressive* — lower-income households pay a larger *share* of income in higher prices.
- Why? Lower-income households spend more on tariffed goods (clothing, food, electronics) as a share of income.
- The 2018 tariffs on consumer goods from China cost US households ~\$800/year on average.
- This regressivity is often ignored in political debates about “protecting workers.”

The Optimal Tariff: Large-Country Case

A large country can lower the world price by restricting imports.

Mechanism:

1. Tariff $\Rightarrow$ Home's import demand falls.
2. Foreign export supply curve slopes up $\Rightarrow$ world price falls to P'_w .
3. Home pays $P'_w + t$ domestically, but $P'_w < P_w$.
4. Net domestic price increase $= t - (P_w - P'_w) < t$.
5. **Part of the tariff burden falls on foreign exporters:** terms of trade improve.

Optimal tariff formula:

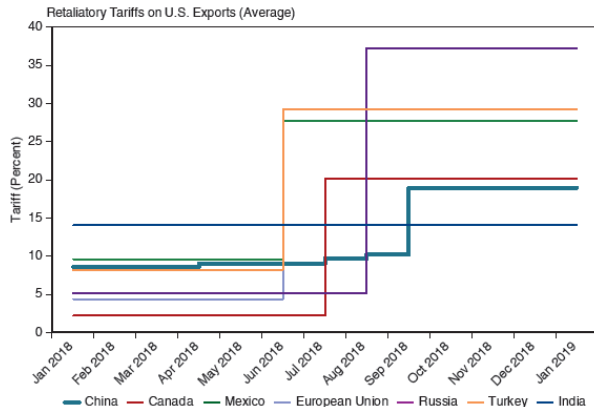
$$t^* = 1$$

The Optimal Tariff Temptation

Every large country has an incentive to impose the optimal tariff — it improves welfare unilaterally by shifting rent from foreigners.

But: if all countries do this simultaneously, a *trade war* erupts and all are worse off. This is the **prisoner's dilemma** of trade policy — the rationale for GATT/WTO binding commitments.

Figure 9-13: Average Retaliatory Tariffs on US Exports (2018)



What this figure shows:

- When the US imposed tariffs in 2018, trading partners (China, EU, Canada, Mexico) retaliated with tariffs on US exports.
- The figure shows average retaliatory tariffs by sector.
- **Targeted retaliation:** foreign countries targeted sectors with political significance in the US (soy beans from Iowa, bourbon from Kentucky, motorcycles from Wisconsin — where Harley-Davidson is headquartered).
- This is strategic retaliation: maximise political pressure on US politicians with minimum economic self-harm.

Import Quotas vs. Tariffs

Import quota: limit on physical quantity of imports.

At equivalent restriction level: quota raises domestic price by same amount as tariff.

Key difference: who captures the scarcity rent?

- **Tariff:** government captures revenue (area c).
- **Quota (auctioned licences):** equivalent to tariff.
- **Quota (free licences):** rent goes to domestic importers or licence holders.
- **Voluntary Export Restraint (VER):** rent accrues to *foreign* exporters — worst outcome for importing country.

VERs: Worse than Tariffs

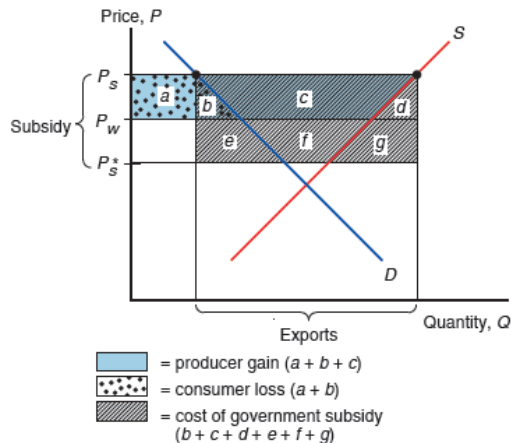
In a VER, foreign exporters *voluntarily* restrict exports (under threat of worse alternatives). They capture the quota rent because their export price rises.

The US used VERs on Japanese cars (1980s): Japanese automakers captured ~\$1B/year in rents. US government got nothing.

Canadian Example

Canada's dairy supply management uses TRQs: tiny quota at 0%, then 298% on butter above. Dairy farmers capture

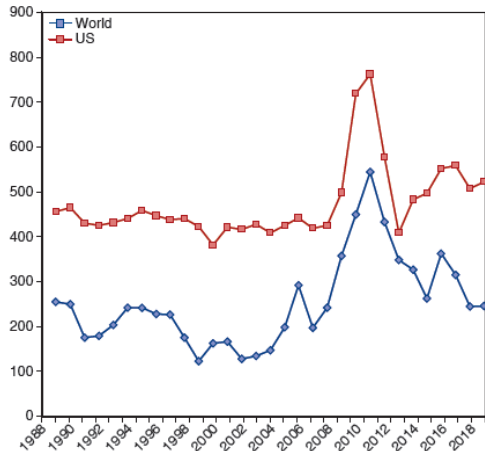
Figure 9-14: Effects of an Export Subsidy



What this figure shows:

- An export subsidy s raises the price that domestic firms receive above the world price.
- Domestic consumers face *higher* prices (less supply available domestically).
- World price *falls* (more supply on world market).
- ToT *deteriorate* for the exporting country.
- Net welfare effect:
 - CS falls (domestic price rises).
 - PS rises (export price rises).
 - Government *pays* the subsidy.
 - Net: negative — worse than a tariff.

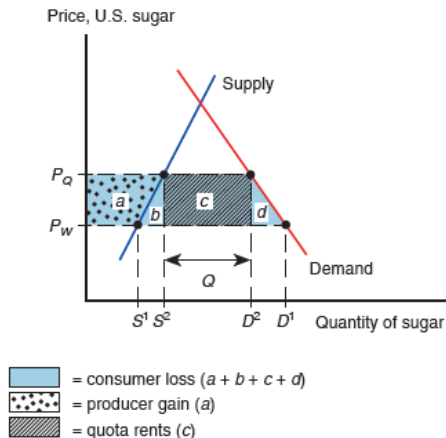
Figure 9-15: US and World Raw Sugar Prices, 1989–2019



Reading Figure 9-15:

- **Blue line:** US raw sugar price (domestic, protected).
- **Red line:** world raw sugar price.
- US sugar price is consistently *above* the world price — the price wedge created by the TRQ.
- The premium fluctuates but has averaged ~50–100% above world price for 30+ years.
- **Who pays?:** US consumers of sugar and sugar-containing products (candy, soda, baked goods).
- **Who gains?:** US sugar cane and sugar beet producers.

Figure 9-16: Effects of the US Import Quota on Sugar



Reading Figure 9-16:

- The US sugar TRQ effectively functions as an import quota.
- **Welfare decomposition:**
 - **CS loss:** large (US consumers pay $\approx 2\times$ world price).
 - **PS gain:** domestic sugar producers gain.
 - **Revenue:** quota rent goes to licence holders (partly foreign exporters with TRQ access).
 - **Net DWL:** substantial.
- Estimated cost to US consumers: $> \$3$ billion/year.
- Estimated cost per job “saved” in US sugar industry: $> \$500,000$.

Summary: Effects of Alternative Trade Policies

Policy	Dom. price	ΔCS	ΔPS	Gov't revenue	Net welfare
Tariff (small)	$P_w + t$	$-(a + b + c + d)$	$+a$	$+c$	$-(b + d)$
Tariff (large)	$P'_w + t$	$-(a + b + c + d)$	$+a$	$+c + e$	$e - (b + d)$
Import quota	$P_w + t^{eq}$	$-(a + b + c + d)$	$+a$	$+c$ (if auctioned)	$-(b + d)$
VER	$P_w + t^{eq}$	$-(a + b + c + d)$	$+a$	0	$-(b + c + d)$
Export subsidy	$P_w + s$	$-(a + b + c)$	$+a + b$	$-(b + c)$	$-(b + c)$

Note: all compared to free trade benchmark; e = ToT gain for large country.

Ranking of Instruments

For an importing country: $\text{Tariff} \geq \text{Quota (auctioned)} > \text{Quota (free)} \gg \text{VER}$.

For welfare: free trade dominates all (absent market failures or large-country effects).

Key Takeaways

1. Tariffs raise domestic price, protect producers, raise government revenue, and harm consumers. **Net DWL** = $b + d$ for a small country.
2. **Large countries** can gain from an optimal tariff (ToT improvement) — but invite retaliation.
3. **Quotas vs. tariffs**: similar price effects but rent allocation differs. VERs are worst (rent to foreigners).
4. **Export subsidies** simultaneously raise domestic prices and worsen the ToT — net welfare is negative.
5. Real cost per job saved by protection is enormous — far exceeds typical manufacturing wages.
6. US 2018 tariffs cost households $\sim \$800/\text{year}$, were regressive, and invited costly retaliation.

Next: Chapter 10 — Political Economy of Trade Policy

If free trade is so clearly better, why does protection persist?